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Moderate-pressure conversion of H_2 and CO_2 to methanol via adsorption enhanced hydrogenation

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HIGHLIGHTS

- A hybrid catalyst/adsorbent consisting of Cu-ZnO-ZrO_2 and hydrotalcite was prepared.
- The catalyst showed elevated uptakes of H_2O and CO_2 compared to adsorbent-free catalyst.
- The enhancing effects of the hydrotalcite support on methanol selectivity were verified.
- A ~61.3% savings in energy consumption for compression was achieved.

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ABSTRACT

CO_2 hydrogenation to methanol plays an increasingly important role in fields of chemical engineering, energy generation and H_2/CO_2 utilization, and the prohibitive costs result partially from the high operating pressures required for practical application. Thus, a hybrid catalyst/adsorbent consisting of Cu-ZnO-ZrO_2 supported on hydrotalcite (named CZZ@HT) was synthesized, characterized and analyzed in this study with the intent that the adsorbent hydrotalcite would enhance the local concentration of CO_2 and assist in catalyst dispersion. The as-prepared CZZ@HT catalyst containing 43.4 wt% of CuO-ZnO-ZrO_2 in the form of well dispersed nanoparticles possessed a considerable BET surface area and external surface area after reduction. A remarkable copper dispersion of 58.7% was thereby achieved. This reduced catalyst displayed elevated uptakes of H_2O and CO_2 at 473 K compared to the reference adsorbent-free catalyst and presented enhanced adsorption capacities of CO_2 at reaction temperatures due to collective effects of physisorption and chemisorption. Catalysis experiments on a fixed bed reactor using the rCZZ@HT catalyst showed a methanol selectivity of 83.4% and a S_{MeOH}/S_{CO} ratio of 5.0 in products. A control experiment in which hydrotalcite was replaced with quartz (named rCZZ&QS) showed considerably lower conversion at low pressure and demonstrated the enhancing effect of the hydrotalcite support. The new catalyst could achieve the same methanol productivity as the control catalyst at 2.45 MPa lower reaction pressure. This lower pressure corresponds to a ~61.3% savings in energy consumption for compression. Accordingly, the CZZ@HT is a promising candidate for CO_2 hydrogenation to methanol at moderate pressures.

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Introduction

Hydrogen carriers such as methane (CH_4), methanol (MeOH), dimethyl ether (DME), etc. are expanding traditional utilization strategies of hydrogen, including CO_2 methanation [1–3], hydrogenation to methanol [4,5], etc. However, converting hydrogen (H_2) and carbon dioxide (CO_2) to methanol remains a challenge industrially at present due to the high costs of production of those chemicals [6,7]. Currently, most methanol synthesis plants use CO or (CO + CO_2) as the carbon source [8,9]. One of the main drawbacks of using CO_2 remains in the boosted hydrogen consumption which elevates total costs of the process. The hydrogen price is expected to be reduced with the progress of novel technologies, but another barrier still lies in the high reaction pressure. Thus, the process of methanol (MeOH) synthesis via CO_2 hydrogenation provides great barriers in application in the emerging carbon capture and utilization (CCU) industry, though it has been highlighted as a future sustainable carbon-neutral energy cycle and for the methanol economy [10,11]. The reason for the difficulty in conversion is rooted in the two competitive reactions which take place simultaneously on most catalysts, viz.

Methanol synthesis (MS)

$$CO_2(g) + 3H_2(g) \leftrightharpoons CH_3OH(g) + H_2O(g)\ \Delta H_0 = -48.17\ \text{kJ/mol} \quad (1)$$

and reverse water gas shift (RWGS)

$$CO_2(g) + H_2(g) \leftrightharpoons CO(g) + H_2O(g)\ \ \Delta H_0 = +41.20\ \text{kJ/mol} \quad (2)$$

In order to elevate CO_2 conversion and suppress CO formation, the reaction is often maintained under strict reaction conditions of high pressure (≥5.0 MPa) in industrial methanol production [9,12]. According to the techno-economic assessment conducted by Pérez-Fortes et al., the cost of the compressors used to elevate the system pressure is 45% of the total equipment cost and the energy consumption of compressing the gas is 66% of the total electricity cost in a typical methanol CCU-plant [6]. Therefore, the modification and development of catalysts have focused on moderate reaction pressures, but the catalytic performance is still unsatisfactory at these conditions, with an average methanol selectivity of only 40–60% at 3.0 MPa and 523 K using general copper-based catalysts (e.g., Cu/ZnO, Cu/Al_2O_3 and Cu/ZnO/Al_2O_3) [13–17].

Novel catalysts have been investigated in the past decades to cope with the above limitations. As reported [18–23], noble metals or bimetallic alloys have been investigated to replace the traditional copper-based catalysts in CO_2 hydrogenation to methanol and remarkable methanol selectivity has been achieved using these catalysts at lowered reaction pressures (usually 3.0 MPa, even under atmospheric pressure and low temperature [24]), which greatly expands the ideas for catalyst design and preparation in this field. Nevertheless, there is still a large gap between results achieved in the laboratory and commercial application, e.g., catalyst stability, catalyst preparation process and cost. There have also been meaningful attempts to perfect the conventional copper-based catalysts by importing special metal elements and improving active sites. The metal elements including zirconium (Zr), manganese (Mn), titanium (Ti), cerium (Ce), etc. which have been used to fine-tune interfaces between their oxides and active copper to promote the stability and methanol selectivity [25–28]. For instance, La-Mn-Cu-Zn-O based catalysts derived from perovskite precursors were reported to improve the reducibility of Cu species for promoted catalytic activity [29]. Cr, Mo and W were also used for modifications of CuO-ZnO-ZrO_2 catalysts to fine-tune basic sites [30]. Among these catalysts, the Cu-ZnO-ZrO_2 catalyst presents outstanding characters in activity and stability, appearing as a promising candidate for industrial usage [31–33]. Some porous supports such as porous silica and activated carbon have also been employed to expand the surface area and dispersion of active copper [34–37].

In our previous study, a novel catalyst consisting of hydrotalcite-like compounds (HTs) and Cu/ZnO/ZrO_2 (CZZ) nanoparticles was synthesized based on the concept of sorption enhancement. HT-based catalysts have been used in hydrogen production systems [38], but the imported HT in CZZ-HT catalyst not only provides considerable surface area to improve copper dispersion, but may also enhance the adsorption of CO_2, leading to a higher local concentration. The methanol selectivity achieved using this catalyst exceeds that of a commercial catalyst within the tested temperature range from 423 K to 573 K. However, adsorption properties of the catalyst at reaction temperatures remain to be analyzed, and its potentials of applications at moderate reaction conditions are not well investigated so far.

Based on our previous work, a highly efficient hybrid Cu/ZnO/ZrO_2 hydrotalcite catalyst (named CZZ@HT) was prepared. The as-synthesized CZZ@HT consists of CZZ nanoparticles and amorphous HT, showing considerable BET surface area, copper surface area and copper dispersion. It was characterized by XRD, SEM, EDX, low-temperature physisorption of N_2, N_2O titration and high-temperature adsorption of H_2O and CO_2. Contributions of hydrotalcite in the catalyst to its CO_2 adsorption properties were clarified based on analysis of CO_2 isotherms on the sample. Its catalytic properties were then systematically analyzed by conducting reaction experiments on a fixed-bed micro reactor. Reaction temperature and space velocity were optimized, and effects of operating pressures on methanol synthesis was investigated and further evaluated from the viewpoint of energy conservation.

Catalyst synthesis and characterization

Materials and reagents

The hydrotalcite (HT) used in this work was the commercial Pural MG50 produced by SASOL, Germany. All HT samples were activated at 673 K for 4 h in air before usage. $Cu(NO_3)_2\cdot 2.5H_2O$, $Zn(NO_3)_2\cdot 6H_2O$ and $ZrO(NO_3)_2\cdot xH_2O$ were supplied by Sigma-Aldrich and K_2CO_3 was supplied by Ajax. Homemade deionized water (DW) was used for solution preparation.

Preparation of CZZ@HT catalyst

CZZ@HT catalyst was synthesized via a modified co-precipitation method. In a typical procedure, 1 M ternary

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metal nitrate solution consisting of 0.6 M copper, 0.3 M zinc and 0.1 M zirconium (solution A) and 1.2 M K_2CO_3 solution (solution B) were prepared with deionized water. The activated HT (6 g) was uniformly dispersed in 25 mL of solution A to obtain a slurry, which was then ultrasonicated for 5 min and stirred for 2 h. Under continuous stirring, solution B was added dropwise to the slurry until the pH reached approximately 9.5. The resultant mixture was further stirred for 2 h, aged for 1 h, filtered and washed with deionized water to remove the residue. After drying at 393 K overnight, the sample (labeled as CZZ@HT) was finally collected and calcined in air at 673 K for 4 h.

CZZ catalyst without HT was also prepared via the conventional co-precipitation method as a reference. Specifically, the aforementioned solution B was slowly added to the solution An under continuous stirring. The mixture was afterwards stirred for 2 h and aged for 1 h, following by filtration, washing and drying at 393 K overnight. The precursor was calcined at 673 for 4 h in air and then the CZZ catalyst sample was collected.

CZZ&QS catalyst was prepared by physically mixing the CZZ catalyst above and quartz sand (QS, Sigma-Aldrich). The CZZ content in CZZ&QS catalyst was 47.6 wt% and was similar to that in CZZ@HT catalyst.

Reduction of catalysts was performed in a tube furnace in 5% H_2 flow (50 mL min^{-1}). The sample was heated with a ramping rate of 5 K min^{-1} and held at 573 K for 6 h. The reduced sample was collected after it cooled to ambient temperature in N_2 flow (50 mL min^{-1}) and were labeled as rCZZ@HT and rCZZ, respectively.

Characterization

Chemical composition of the CZZ@HT sample was analyzed on an inductively coupled plasma emission spectrometer (ICP-OES. Varian 720 ES). Microstructures were tested by the Powder X-ray diffraction (PXRD. PW1140/90, Phillips Analytical) equipped with the Cu-Kα radiation. Morphologies and metal dispersions were determined by scanning electron microscope (SEM) and energy dispersive spectroscopy (EDX), respectively (JSM-7001F, JEOL).

Brunauer-Emmett-Teller surface area (BET SA) and density functional theory pore size distribution (DFT PSD) were calculated with N_2 adsorption isotherms at 77 K (ASAP 2010, Micromeritics). Before experiments, the sample was vacuum degassed at 573 K overnight to remove adsorbed guest molecules.

Copper surface area (Cu SA) and copper dispersion (Cu D) were measured via the N_2O titration method (BELCAT-M, MicrotracBEL). Specifically, the sample was installed in a quartz U-tube and degassed under flowing Argon at 573 K for 3 h. When cooled to 303 K, it was reheated and reduced in-situ in 5% H_2 (balanced with Ar) at 573 K for 6 h. The temperature, after reduction, was controlled at 333 K and 5% N_2O (balanced with He) was pulsed in for 1 h so that the surface metal copper (Cu^0) was oxidized to cuprous oxide (Cu_2O). Finally, 5% H_2 was purged to the sample for reduction where the temperature was slowly elevated from 333 K to 773 K at a ramp rate of 5 K min^{-1}. Concentrations of H_2, CO_2 and H_2O in the outlet were recorded every second by mass spectrometry (BELMass, MicrotracBEL), and the parameters were calculated by assuming a Cu/N_2O stoichiometry of 2 and an atomic copper surface density of 1.46×10^{-19} atoms/m^2 [36,39,40].

Analysis of adsorption properties

Adsorption capacities of both H_2O and CO_2 were analyzed in this work (Belsorp-Max, MicrotracBEL).

The adsorption isotherms of H_2O on rCZZ@HT and HT were measured at 473 K which was within the investigated reaction temperature range of 423–573 K. In a typical procedure, 0.1 g of sample was loaded in a sample tube, vacuum degassed at 623 K for 10 h to remove adsorbed impurities and weighted to calculate the weight loss after degas. The tube was then installed in the measurement apparatus. After it was heated to 473 K by a heater and vacuumed to a pressure below 10^{-3} Pa, a small quantity of H_2O vapor (~0.3 mL) was pulsed into the system each time so that the value of equilibrium pressure could be measured and recorded (pressure variation$\leq$0.1% within 3600 s). The adsorption isotherm was thereby obtained in the range of 0–2.5 kPa by intermittent pulsing of H_2O vapor.

Adsorption isotherms of CO_2 on rCZZ@HT, HT and rCZZ were also measured at 473 K, and for comparison those on the CZZ@HT was further analyzed at 423 K and 523 K. Before each measurement, the sample was vacuum degassed at 623 K for 10 h. The sample, filled in a sample holder installed in the apparatus, was preserved initially at a set temperature at a vacuum pressure of $\leq$0.01 Pa, and a fixed amount of CO_2 (0.3–0.5 mL) was pulsed into the system each time. The pressure was recorded when it reached equilibrium an equilibrium state. An adsorption isotherm was attained by repeating the process above until the pressure came up to ~100 kPa.

As CO_2 uptakes on rCZZ@HT were collective effects of physisorption and chemisorption, the model combining classical single-site Langmuir and chemisorption-surface reaction [41–43] equations was employed in this study for analysis of CO2 adsorption isotherms. It is given by

$$n(P,T) = \frac{MK_LP}{1+K_LP} + \frac{mK_CP[1+(a+1)K_RP^a]}{1+K_CP+K_CK_RP^{(a+1)}} \tag{3}$$

where n (mmol g^{-1}) is the equilibrium adsorption capacity of CO_2 at the temperature T (K) and pressure P (kPa), M (mmol g^{-1}) is the saturated physisorption amount of CO_2 on the surface of the catalyst, m (mmol g^{-1}) is the saturation chemisorption capacity of CO_2 on the surface of the catalyst, K_L (kPa^{-1}) is the equilibrium constant for physisorption, K_C (kPa^{-1}) is the equilibrium constant for chemisorption reaction, K_R (kPa^{-a}) is the equilibrium constant for additional complexation reaction, and a is the stoichiometric coefficient for complexation reaction (an exponential function of temperature). Thermodynamic constants of K_L, K_C and K_R are functions of temperature:

$$K_L = K_L^0 \exp\left(\frac{Q_{st}}{RT}\right) \tag{4}$$

$$K_C = K_C^0 \exp\left(\frac{q_C}{RT}\right) \tag{5}$$

$$K_R = K_R^0 \exp\left(\frac{\Delta H_R}{RT}\right) \tag{6}$$

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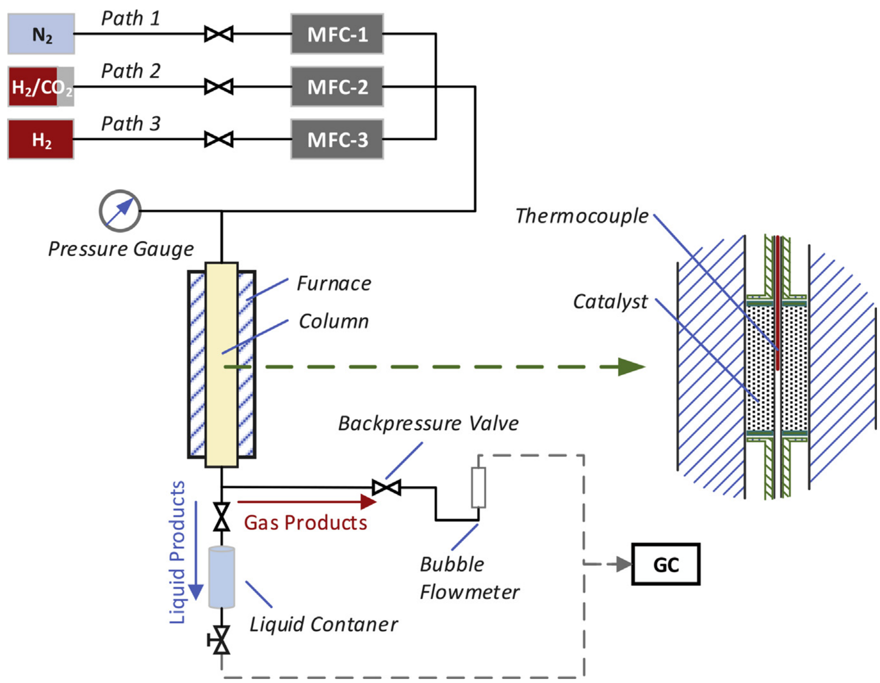

Fig. 1 – Schematic of the fixed bed micro reactor.

where Q_{st} (kJ mol^{-1}) is the isosteric heat of physisorption, q_c (kJ mol^{-1}) is the isosteric heat of chemisorption, ΔH_R (kJ mol^{-1}) is the heat of additional surface reaction, K_L^0 (kPa^{-1}), K_C^0 (kPa^{-1}) and K_R^0 (kPa^{-a}) are constants.

Catalytic reaction studies

Fixed bed micro reactor

A fixed bed micro reactor was used for the analysis of catalytic performance. Herein, three inlet gas paths equipped with mass flow controllers (MFC) were designed for N_2, H_2 and mixed gas of H_2/CO_2, respectively, and the reaction column with the dimensions of Ø10 mm × L 500 mm was placed in a three-stage furnace. The pressure in the system was maintained with a back pressure regulator. The reaction temperature, measured by a thermocouple, was well controlled as the height of catalyst (1 g) positioned in the center section of column was only about 40 mm (occupying only 8% of the total column length). The gas passed through the system controlled by a MFC and the catalytic reaction was conducted within the catalyst bed. Off gases after the reaction would flow into a condenser where liquid products and gas products were separated, and then the liquid products were collected in the connected container and gas products emitted and collected after the back pressure regulator. Both liquid and gas samples were analyzed on the gas chromatography (GC7890B, Agilent) equipped with thermal conductivity detector (TCD) and flame ionization detector (FID). The outlet flowrate of gas products was measured with a bubble flowmeter located at the outlet in the system. Fig. 1 presents the schematic of this micro reactor briefly.

Optimization of reaction conditions

Reaction conditions of temperature and space velocity were optimized prior to experiments of varying reaction pressures.

The reaction temperature was investigated within the range from 423 to 573 K (with an interval of 50 K). In a typical run, uniformly mixed particles of 1 g catalyst (50–70 mesh) and 1 g quartz sand (50–70 mesh, Sigma-Aldrich) were positioned in the center section of the column, where they were reduced in-situ in pure H_2 flow at 573 K for 6 h (Path 2) and then naturally cooled to ambient temperature in N_2 flow (Path 1). Afterwards, the inlet was switched to the mixed gas of H_2/CO_2 (Path 3, 74.98% H_2 and 25.02% CO_2, BOC). The pressure of 3.0 MPa and space velocity of ~4000 (±1.1%) mL $g_{catalyst}^{-1}$ h^{-1} were set by adjusting the back pressure and the mass flow controller (MFC-2), respectively. The reaction temperature was measured with the thermocouple placed into the center of the catalyst and controlled by the system temperature controller. Flowrates and components of off-gas were measured every 2 h until the reaction reached equilibrium, and liquid products were analyzed after the reaction. Mass and volume of catalyst were measured again for reference at the end of all experiments. The optimal reaction temperature for this catalyst could be determined by comparing the methanol STY and yield.

To determine a suitable space velocity of H_2/CO_2 mixed gas used for the subsequent moderate-pressure reactions, experiments with different space velocities were conducted at the optimal reaction temperature. Four space velocities were investigated using the procedure described above, including 1350, 2700, 4050 and 5400 mL $g_{catalyst}^{-1}$ h^{-1} (±0.12%).

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Catalysis experiments at moderate operating pressures

Both CZZ@HT (CO_2 adsorbing hydrotalcite support) and CZZ&QS (non-adsorbing quartz support) were evaluated at moderate pressures. The reactions were performed in the pressure range from 0.5 to 3.0 MPa.

Calculation of catalytic performance

To examine the catalytic properties of the samples, indicators including CO_2 conversion (X_{CO2}, %), methanol selectivity (S_{MeOH}, %), CO selectivity (S_{CO}, %), methanol space time yield (STY, $g_{MeOH}\ kg_{catalyst}^{-1}\ h^{-1}$) and yield (Y, %) were calculated as

$$X_{CO_2} = \left[1 - \frac{F_{out} \times p_{out,CO_2}}{F_{in} \times p_{in,CO_2}}\right] \times 100\% \tag{7}$$

$$S_{MeOH} = \left[1 - \frac{F_{out} \times \left(p_{out,CO} + p_{out,CH_4}\right)}{F_{in} \times p_{in,CO_2} - F_{out} \times p_{out,CO_2}}\right] \times 100\% \tag{8}$$

$$S_{CO} = \frac{F_{out} \times p_{out,CO}}{F_{in} \times p_{in,CO_2} - F_{out} \times p_{out,CO_2}} \times 100\% \tag{9}$$

$$STY_{MeOH} = \frac{m_{MeOH}}{m_{catalyst} \times t} \tag{10}$$

$$Y = X_{CO2} \times S_{MeOH} \times 100\% \tag{11}$$

where F_{in} (NmL min^{-1}) was the inlet flowrate, F_{out} (NmL min^{-1}) was the outlet flowrate, $p_{in,CO2}$ (%) was the molar percentage of CO_2 in the feed gas, $p_{out,CO2}$, p_{CO} and p_{CH4} were molar percentages (%) of CO_2, CO and CH_4 in off-gas; m_{MeOH} (g) was the mass of methanol produced during the reaction, $m_{catalyst}$ (g) was the mass of catalyst, and t (h) was the period of reaction. The observed carbon-containing products in both gas and liquid phase only appeared as methanol, CO and CH_4 in all experiments, thus, the carbon balance could be checked from the equations above and confirmed within ±3%.

Results and discussion

Textural properties

Structures and morphologies

As shown in Table 1, the CZZ@HT catalyst contains metal elements of Mg, Al, Cu, Zn and Zr. The Mg and Al derive from HT, whereas the Cu, Zn and Zr form particles of CZZ. The Cu:Zn:Zr molar ratio is 3.9:2.3:1 with a total CZZ content of 43.4 wt% (calculated as the form of CuO-ZnO-ZrO_2).

The XRD pattern presented in Fig. 2 reveals crystal structures in the CZZ@HT catalyst. Consistent with conventional CZZ catalysts, the CZZ@HT shows strong diffraction peaks of CuO (JCPDS 45-0937) and ZnO (JCPDS 36-1451) with hidden peaks of tetragonal ZrO_2 (JCPDS 50-1089), identifying the formations of CuO and ZnO which are important compositions in the catalyst and can be applied to methanol synthesis from CO_2 hydrogenation after reduction. The CZZ@HT also exhibits several broad peaks of amorphous HT. As there is no peak of Mg/Al spinel appearing in the XRD pattern, the micro structures of HT are retained.

Fig. 3 displays morphologies of the rCZZ@HT with its SEM and EDX images. Fig. 3(a) shows well dispersed CZZ nanoparticles on the external surfaces of the sample. The EDX image in Fig. 3 (b) further describes distributions of the elements of Cu, Zn, Mg, Al and O, where no bright regions from metal agglomeration can be found.

Generally, high surface area will benefit exposure of active sites. The BET surface area of rCZZ@HT reaches 130.4 $m^2\ g^{-1}$ (Table 1), which is much higher than conventional CZZ catalysts reported in the literature [44–46]. Thus, more active copper tends to expose to reactants for better catalytic performance. The as-prepared catalyst, moreover, possesses a remarkably high external surface area of 117.9 $m^2\ g^{-1}$, contributing to 90.4% of its total surface area. From Fig. 4 (pore

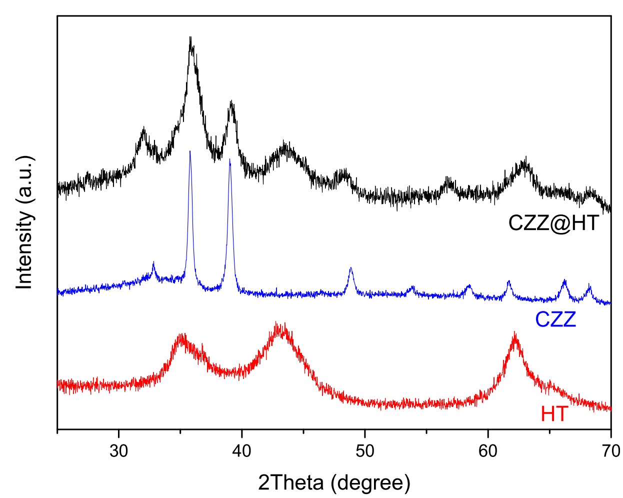

Fig. 2 – XRD pattern of CZZ@HT, CZZ and HT.

Table 1 – Chemical components of CZZ@HT and pore parameters of rCZZ@HT.

	Content (wt. %)					BET SA ($m^2\ g^{-1}$)	External SA ($m^2\ g^{-1}$)	Average pore diameter (nm)
	Mg	Al	Cu	Zn	Zr			
CZZ@HT	14.0	17.0	17.7	10.9	6.6	130.4	117.9	11.6
CZZ	–	–	42.5	31.4	6.2	35.8	12.3	32.8
HT	26.9	29.2	–	–	–	180.0	113.7	3.4

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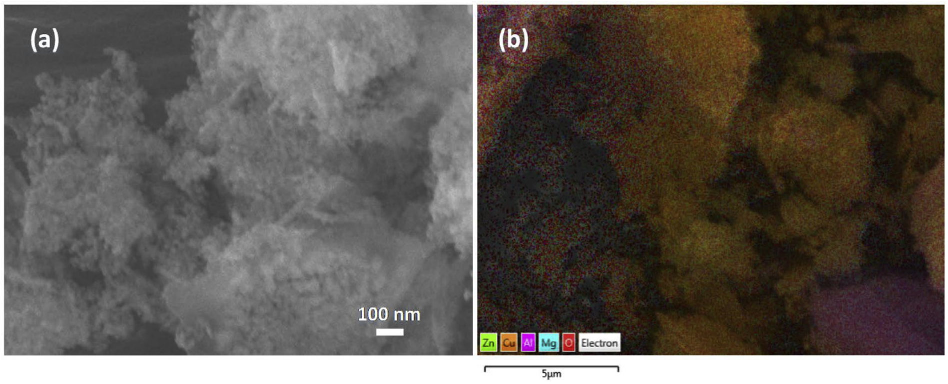

Fig. 3 – SEM image (a) and EDX mapping (b) of rCZZ@HT.

size distribution), the sample rCZZ@HT mainly includes mesopores and macropores, which reveals that the pore systems in the catalyst sample not only inherit micropores and mesopores from HT, but also develop a large amount of macropores formed from co-precipitated CZZ nanoparticles.

The characteristics observed above would benefit dispersion of active copper and hence we expected enhanced catalytic performance.

Surface active sites

As the surface copper plays a crucial role in CO_2 hydrogenation to methanol, copper surface areas and copper dispersions of as-prepared rCZZ@HT and reference rCZZ were measured and are presented in Table 2. Though copper surface areas per gram catalyst of both samples are fairly close, the copper surface area per gram CZZ in rCZZ@HT (only considering active components) is over twice as much as that in the rCZZ, while the corresponding copper dispersion in rCZZ@HT is also 2.6 times of that in rCZZ. Thus, the HT as supporting material significantly promotes the copper dispersion in rCZZ@HT. Since the surface copper acts as the active site in rCZZ@HT, we expect the rCZZ@HT catalyst would be efficiently used in the catalytic reaction for methanol synthesis from CO_2.

Adsorption capacities for H_2O and CO_2 of HT and prepared catalysts

Due to very weak surface interactions, the adsorption of H_2 does take place at reaction temperatures within the range

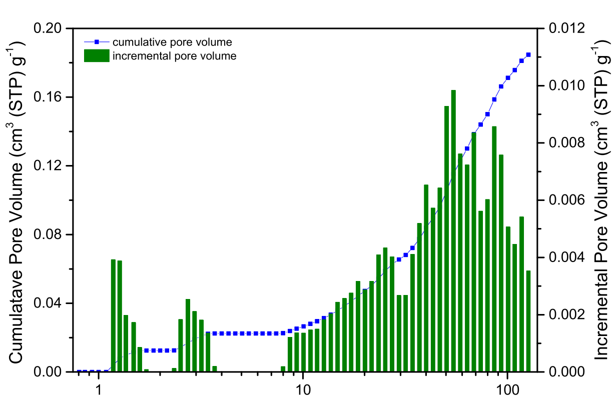

Fig. 4 – DFT pore size distribution of rCZZ@HT.

423–573 K [47]. Alternatively, since CO_2 and H_2O are reagent and product in both MS and RGWS reactions, it is important to investigate adsorption capacities of catalysts for both H_2O and CO_2. The reduced rCZZ sample itself has a low BET surface area of 33.8 m^2 g^{-1}, so it is a very weak adsorbent. However, the incorporated HT in rCZZ@HT would improve the adsorption capacities of the catalyst for H_2O and CO_2. At a temperature of 473 K which is within the real reaction temperature range (usually 453–573 K), the adsorption capacities of both H_2O and CO_2 were measured and are shown in Fig. 5. Because the rCZZ shows negligible H_2O adsorption property as most metal oxides do, it is not presented in this figure. The H_2O adsorption amounts on rCZZ@HT are 40% of those of HT on average, correlating to the HT ratio in the rCZZ@HT sample (Fig. 5(a)). H_2O adsorbed on active sites of catalysts prohibits the methanol synthesis significantly, so that H_2O molecules generated in reactions should be quickly removed to avoid further RWGS reaction [48]. Thus, suitable flowrates of inlet gas are important to purge the H_2O for promoting CO_2 conversions (increase the external mass transfer coefficient).

CO_2 adsorption properties are improved proportionally as HT is incorporated into the catalyst. As shown in Fig. 5 (b), the rCZZ itself adsorbs only a small amount of CO_2 (about 0.036 mmol g^{-1}). Though it is nearly nonporous in microstructure, its basic sites originating in ZnO/ZrO_2 compositions play crucial roles in bonding CO_2, appearing as detectable chemisorption. The HT, by comparison, exhibits an order of magnitude higher CO_2 uptake than the rCZZ because of its layered structures and rich basic sites. As a co-contribution of CZZ and HT, the CZZ@HT possesses intermediate CO_2 adsorption capacities. Adsorption capacities of CO_2 increase rapidly at low pressures and raise slowly when the pressure further elevates.

Isotherms of CO_2 adsorption on the rCZZ@HT can be well fit by the derived mathematic model (Eq. (3)) presented in this

Table 2 – Copper surface areas and copper dispersions of rCZZ@HT and rCZZ.

	Copper surface area (m^2 $g_{catalyst}^{-1}$)	Copper surface area (m^2 g_{CZZ}^{-1})	Copper dispersion (%)
rCZZ@HT	48.2	106.8	58.7
rCZZ	50.8	50.8	22.9

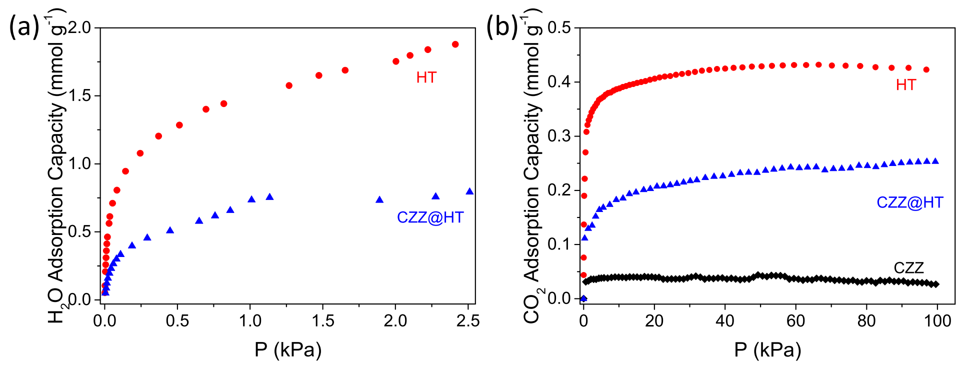

Fig. 5 – Adsorption isotherms of (a) H_2O and (b) CO_2 at 473 K (circle-HT; triangle-reduced CZZ@HT).

work with a high correlation coefficient of 0.998 (Fig. 6), and the obtained parameters for the model are shown in Table 3. Q_{st} is the physisorption adsorption heat and its value is similar to those of similar physisorbents [49,50]. Moreover, the value of q_C and ΔH_R reveal strong interactions between CO_2 and active sites on rCZZ@HT. Because q_C and ΔH_R of CZZ@HT are about 3 times and 2 times of those measured on K_2CO_3 promoted hydrotalcite [42], enhancements in chemisorption heat and surface reaction heat should result from the CZZ which helps promote CO_2 activation during reactions. Thus, the considerable CO_2 uptakes of CZZ@HT at reaction conditions may increase the CO_2 concentration immediately adjacent to the active sites, which may enhance catalytic performances of the CZZ@HT catalyst through surface diffusion of the CO_2. Ideally, physisorption, chemisorption and chemical complexation take place simultaneously on CZZ@HT [41]. As the adsorption temperature increases from 423 to 523 K, both physisorption and chemisorption are weakened, leading to decreased total CO_2 adsorption capacities on the rCZZ@HT. The analysis from the mathematic model demonstrates that the ratio of physisorption to total CO_2 adsorption capacities at the pressure of 100 kPa changes from 31.3% at 423 K to 18.6% at 523 K.

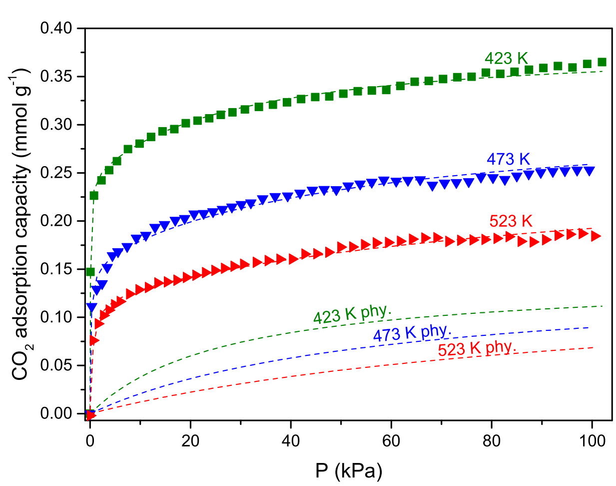

Fig. 6 – CO_2 adsorption isotherms of rCZZ@HT at 423, 473 and 523 K (points-measured data, dash line-simulated isotherm).

Catalytic performance

Optimization of reaction temperature and space velocity
Catalytic activity of the CZZ@HT sample in reactions is strongly influenced by reaction temperature. As shown in Fig. 7 (a), when the reaction temperature is raised from 423 K to 573 K, the overall CO_2 conversion increases from 2.6 to 5.7%. Specifically, the fraction of CO_2 converted to CO continually increases with elevated temperature as the endothermic RWGS reaction is enhanced thermodynamically at high temperatures. The conversion of CO_2 into methanol, in contrast, reaches a peak at 523 K and then decreases as the temperature is further elevated. The methanol synthesis reaction is exothermic and its equilibrium extent decreases at higher temperatures. The temperature of 523 K is the optimal point for the balance of catalytic activity where the highest CO_2 conversion to methanol of 4.5% is achieved. As shown in Fig. 7 (b), the selectivity of methanol in products shows inverse correlation to the reaction temperature, resulting from both reduced methanol synthesis and enhanced CO generation. Surprisingly, a methanol selectivity of 83.4% is achieved with the CZZ@HT at 523 K and 3.0 MPa with a S_{MeOH}/S_{CO} ratio of 5.0, while that of CZZ measured at the same condition is only 68.0% in our previous experiments [51]. From Fig. 7 (c), low temperature benefits the ratio of S_{MeOH}/S_{CO}, however, the low CO_2 conversion at a low temperature results in lower methanol product. Fig. 7 (d) displays more apparent comparisons of methanol STY and yield, in which both parameters achieve their peak values with STY of 64.5 $g_{methanol}\ kg_{catalyst}^{-1}\ h^{-1}$ and yield of 4.5% at 523 K. Therefore, the optimal reaction temperature of 523 K was applied to the subsequent experiments.

Five space velocities were investigated to determine the optimal value for this catalyst in methanol synthesis. As shown in Fig. 8 (a), the overall CO_2 conversion gradually decreases along with increasing space velocity because of the shortened residence time of reactants in the catalyst bed. Because of the reduction in residence time, CO_2 conversion to CO and methanol, respectively, results in declined trends of CO and methanol concentration. However, at low space velocity (2000 mL $g_{catalyst}^{-1}\ h^{-1}$), the conversion of CO_2 to methanol shows the lowest value. We attribute this to poor external mass transfer of the water from the active sites on the surface of the catalyst by gas flow so that the production of methanol is suppressed [52–54]. The derived selectivity of methanol, CO

Table 3 – Parameters of adsorption model for CO_2 adsorption on rCZZ@HT.

T (K)	K_L (10^{-2} kPa^{-1})	K_C (kPa^{-1})	K_R (kPa^{-a})	a	M (mmol g^{-1})	Q_{st} (kJ mol^{-1})	m (mmol g^{-1})	q_C (kJ mol^{-1})	ΔH_R (kJ mol^{-1})
423	3.67	65.11	10.00	1.10	0.141	25.12	0.116	63.53	85.05
473	1.73	9.66	0.78	0.52					
523	0.94	2.06	0.10	0.28					

and the preferred ratio of selectivity (S_{MeOH}/S_{CO}) are shown in Fig. 8(b) and (c), and STY and yield of methanol in Fig. 8 (d). It is observed that the selectivity of methanol increases but the selectivity of CO decreases as space velocity increases, resulting in both preferred ratio and STY of methanol increasing with the increased space velocity. However, the methanol yield tends to peak at approximate 4.48% when the space velocity reaches 2700–4000 mL $g_{catalyst}^{-1}$ h^{-1}. In consideration of methanol selectivity, yield and STY, the optimal space velocity of ~4000 mL $g_{catalyst}^{-1}$ h^{-1} is adopted in the remaining experiments.

Profile of catalytic performance at various reaction pressures

One of the goals of this study was to achieve enhanced performance at lower operating pressures. For this reason, we studied the effect of lower pressures with our catalyst. Profiles of catalytic performance at various reaction pressures for rCZZ@HT and rCZZ catalysts are shown in Fig. 9 (a) and 9 (b). The CZZ@HT still offer good catalytic properties for CO_2 hydrogenation to methanol at low reaction pressures. The overall CO_2 conversion enhances from 4.1% to 5.4%, the CO_2 converted to methanol increases from 2.9% to 4.5% and that to CO decreased from 1.1% to 0.9% when the reaction pressure is elevated from 0.55 MPa to 2.97 MPa. This trade-off in CO_2 conversion reflects the crucial role of reaction pressure in promoting methanol synthesis. However, the CO_2 conversion using rCZZ&QS as catalyst changes very slightly with the reaction pressure over this range. CO_2 converted to methanol slightly increases from 2.4% to 2.8% while that to CO decreased from 1.1% to 0.8%. Although CZZ contents in both samples are similar, the Cu surface area of rCZZ&QS is only about 22.7 m^2 $g_{catalyst}^{-1}$ when QS is considered as an inert catalyst and that of rCZZ@HT is 48.2 m^2 $g_{catalyst}^{-1}$. The rCZZ@HT presents more than twice the copper dispersion as the rCZZ&QS sample. Thus, favorable characteristics in rCZZ@HT contribute to better catalytic performance. Specifically, the adsorption enhancement facilitates methanol selectivity. From the viewpoint of product selectivity as shown in Fig. 9 (c), rCZZ@HT gives a methanol selectivity from 71.8% to 83.4%, but rCZZ&QS only shows selectivity from 67.1% to 77.1% as the reaction pressure increases from 0.55 to 3.0 MPa. The improved methanol selectivity with rCZZ@HT reflects better promoted synthesis of methanol, which correlates to adsorption capacities of CO_2 directly. Because the adsorption of CO_2 is enhanced on rCZZ@HT, a CO_2 rich region (the adsorbed phase [55]) appears near the surface of rCZZ@HT where the a high local CO_2 concentration is present. Since CZZ nanoparticles reside on the HT support, the local CO_2 density around the active sites of CZZ is also higher than that in the gas. The synthesis of methanol is therefore facilitated due to

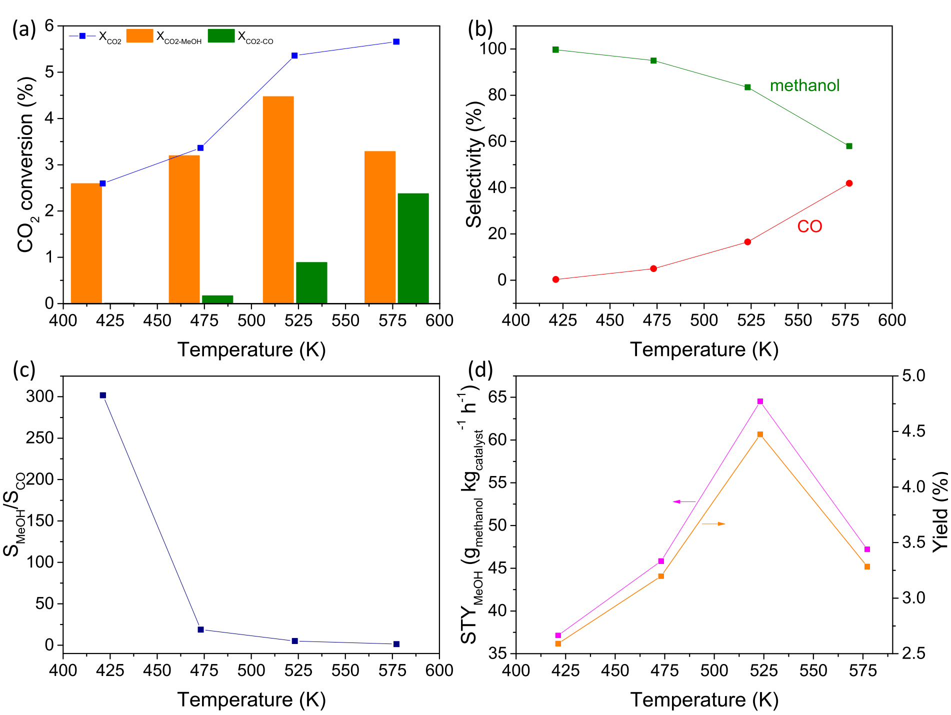

Fig. 7 – Catalytic properties of CZZ@HT at different reaction temperatures. (a) CO_2 conversion, (b) selectivity, (c) S_{MeOH}/S_{CO} and (d) methanol STY and yield (Reaction pressure: 3.0 MPa, space velocity: ~4000 (±1.1%) mL $g_{catalyst}^{-1}$ h^{-1}).

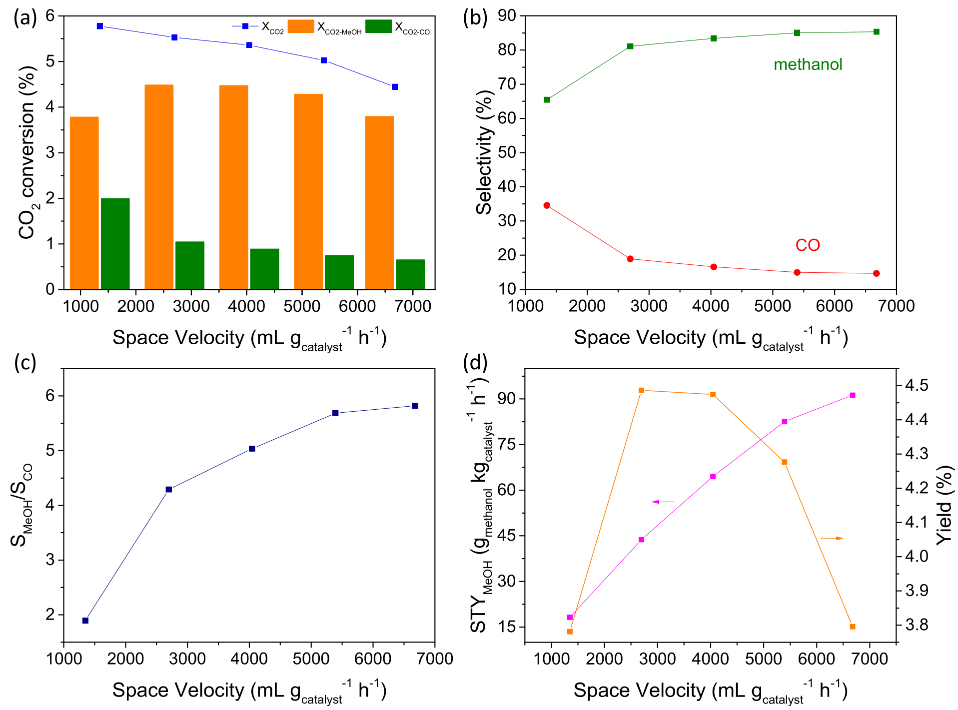

Fig. 8 – Catalytic properties of CZZ@HT at different space velocities. (a) CO_2 conversion, (b) selectivity, (c) S_{MeOH}/S_{CO} and (d) methanol STY and yield (Reaction temperature: 523 K, pressure: 3.0 MPa).

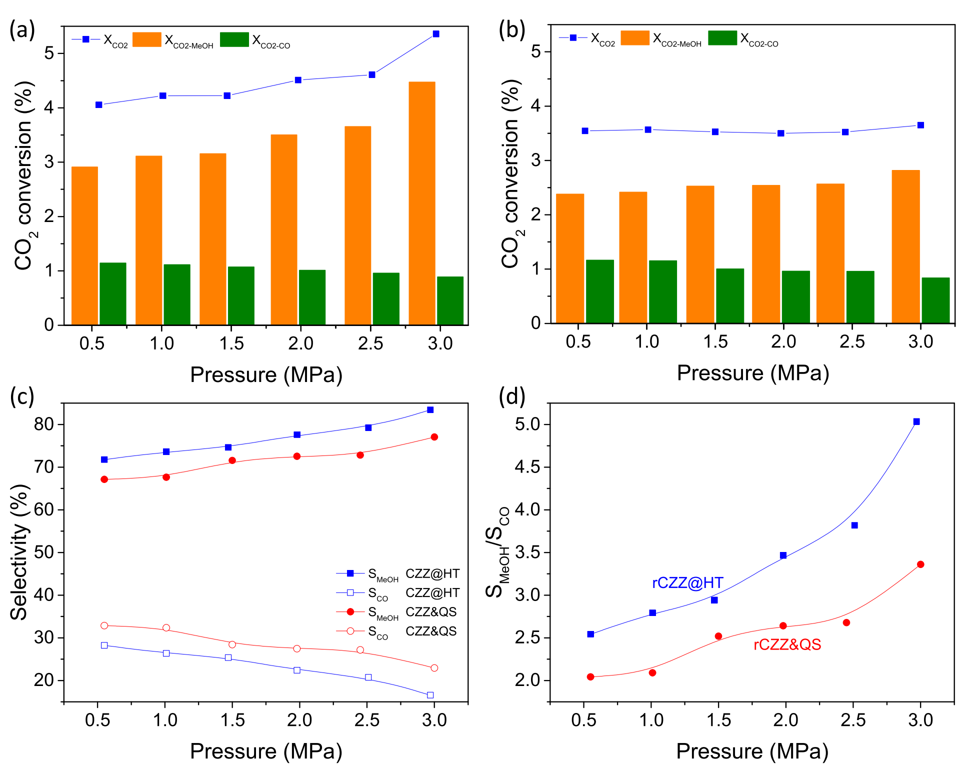

Fig. 9 – Catalytic property summary of CZZ@HT at different reaction pressures. (a) CO_2 conversion of CZZ@HT, (b) CO_2 conversion of CZZ&QS, (c) selectivity and (d) S_{MeOH}/S_{CO} (Reaction temperature: 523 K, space velocity: ~4000 (±1.1%) mL $g_{catalyst}^{-1}$ h^{-1}).

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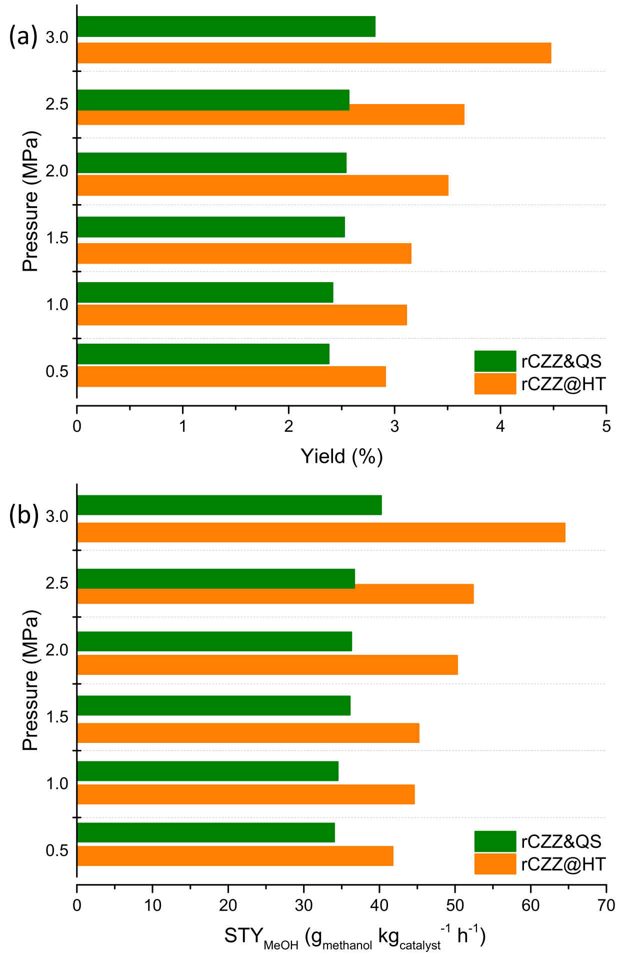

Fig. 10 – (a) methanol yield and (b) STY of rCZZ&QS and rCZZ@HT at different reaction pressures (Reaction temperature: 523 K, space velocity: ~4000 (±1.1%) mL $g_{catalyst}^{-1}$ h^{-1}).

its reduced molecular number after reaction, leading to higher methanol selectivity than rCZZ&QS. Fig. 9 (d) again shows favorable S_{MeOH}/S_{CO} ratios under rCZZ@HT and rCZZ&QS within the entire tested pressure range, showing that the S_{MeOH}/S_{CO} with rCZZ@HT is 1.24–1.50 times that with rCZZ&QS.

Yields and space time yields of methanol productivity achieved by rCZZ@HT and rCZZ&QS are calculated and shown in Fig. 10. Both yields and STYs for the two catalysts increase along with the elevated reaction pressure, but the values for rCZZ&HT are obviously greater than those for rCZZ&QS. The rCZZ@HT shows a methanol yield of 2.91% and a methanol STY of 41.8 g_{MeOH} $kg_{catalyst}^{-1}$ h^{-1} at 0.55 MPa, which exceed those with rCZZ&QS acquired even at 3.0 MPa. Compared to general rCZZ in the tested pressure range, accordingly, the catalytic performance (in consideration of methanol yields and STYs) using the rCZZ@HT is equivalent to that using same amount rCZZ&QS at the elevated pressure of at least 2.5 MPa.

Energy conservation potential using CZZ@HT catalyst

The difference in reaction pressure between CZZ@HT and CZZ/CZZ&QS catalysts to provide similar performance will bring considerable energy savings during operation. Herein, case studies were performed assuming the feed-gas compression as a isentropic process. The energy consumption (kJ/kg) can be evaluated as

$$w_{comp,in} = \frac{kR(T_2 - T_1)}{k-1} = \frac{kRT_1}{k-1}\left[\left(\frac{P_2}{P_1}\right)^{(k-1)/k} - 1\right] \quad (12)$$

where T_1 (K) is the initial temperature of feed-gas, T_2 (K) is the temperature of compressed gas, P_1 (MPa) is the initial pressure of feed-gas, P_2 (MPa) is the target pressure for reaction, R is the gas constant (8.314472 J mol^{-1} K^{-1}), k is the specific heat ratio of mixed gas, i.e., 1.376 (75% H_2 and 25% CO_2) at 293.15 K. Identical initial conditions are fixed (feed-gas pressure: 0.1 MPa, compression temperature: 293.15 K) and results are summarized in Table 4.

In a case where pure CZZ and CZZ@HT are considered, 2.0 and 3.0 MPa was required by CZZ@HT and CZZ, respectively, to guarantee a similar methanol STY of ~50 g_{MeOH} $kg_{catalyst}^{-1}$ h^{-1}. As a result of the reduction in pressure of 1.0 MPa using CZZ@HT catalyst, approximately 17.3% of the energy consumption for gas compression can be saved. It is worth noting that the CZZ@HT only contains 43.4 wt% of CZZ. If the reference catalyst (viz. CZZ&QS) possess equal CZZ contents to CZZ@HT, the energy conservation of CZZ@HT is even more significant. The highest methanol STY realized by the CZZ&QS is ~40 g_{MeOH} $kg_{catalyst}^{-1}$ h^{-1} at 3.0 MPa, which is similar to that for the CZZ@HT at 0.55 MPa. As a consequence, the CZZ@HT will save energy by up to

Table 4 – Comparisons of energy cost using CZZ, CZZ&QS and CZZ@HT catalysts.

Catalyst	CZZ content (wt.%)[a]	Target STY (g_{MeOH} $kg_{catalyst}^{-1}$ h^{-1})	Feed-gas pressure (MPa)	Required reaction pressure (MPa)	Saved energy (%)[b]
CZZ	100	~50	0.1	~3.0	~17.3
CZZ@HT	43.4	~50	0.1	~2.0	
CZZ&QS	47.6	~40	0.1	~3.0	≥61.3
CZZ@HT	43.4	~40	0.1	≤0.55	

[a] CuO-ZnO-ZrO_2 content in catalyst before reduction.
[b] Energy saved using CZZ@HT during the feed-gas compression.

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~61.3%. Thus, CO_2 hydrogenation to methanol in moderate pressure using the novel CZZ@HT catalyst can be realized.

Conclusion

A hybrid catalyst/adsorbent CZZ@HT catalyst combining CZZ and HT was prepared in this investigation. The as-obtained catalyst contains 43.4 wt % of CuO-ZnO-ZrO_2 in the form of crystal CZZ metal oxides on the amorphous hydrotalcite. Well dispersed CZZ nanoparticles are observed on the external surfaces of CZZ@HT without agglomeration producing a BET surface area of 130.4 $m^2\ g^{-1}$ with 90.4% of external surface, contributing to a high copper dispersion of 58.7%. Adsorption capacities of H_2O and CO_2 on the CZZ@HT catalyst are also improved remarkably because of the joint effects of physisorption and chemisorption of the HT. Catalytic experiments were conducted to determine the optimum reaction temperature of 523 K and space velocity of ~4000 mL $g^{-1}_{catalyst}\ h^{-1}$ for the CO_2 hydrogenation to methanol using the CZZ@HT catalyst, in which the maximum S_{MeOH}/S_{CO} ratio in products is achieved (5.0 at the reaction pressure of 3.0 MPa). The CZZ@HT maintains favorable methanol selectivity and thus S_{MeOH}/S_{CO} ratio in all experiments, even at a low reaction pressure of 0.55 MPa. To produce similar methanol STY, the CZZ@HT requires about 2.45 MPa lower reaction pressure than physically mixed CZZ&QS, and thus saves ~61.3% energy consumption for feed-gas compression. Therefore, the CZZ@HT catalyst offers promise for CO_2 hydrogenation to methanol at moderate pressures, which is important for the commercialization of such technology.

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